1.b

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{d} \beta_i \theta_i^2 \quad , \ \beta_i > 0 \text{ constant scalar}, \ \beta_i \in \mathbb{R}$$

$$\nabla J(\theta) = \frac{1}{\cancel{2}} \cdot \cancel{2} B\theta = B\theta.$$

the gradiant descent rule

$$\theta_i^{[t]} = \theta_i^{[t-1]} - \alpha_i \beta_i \theta_i^{[t-1]}$$

$$\theta_i^{[t]} = (1 - \alpha\beta_i)\, \theta_i^{[t-1]}$$

for $\theta_i^{[t]}$ to converge at minimum,

$$|1 - \alpha\beta_i| < 1$$

$$-1 < 1 - \alpha\beta_i < 1$$

leading to $0 < \alpha < 2/\beta_i$

when GD Converges, the gradiant should be zero

ie $\nabla J(\theta^+) = 0$

ie $\beta_i \theta_i^+ = 0$, Since all the $\beta_i > 0$,

the value of θ_i^+ at convergence is zero

$\theta^+ = 0$